

LLM-BASED TOXICITY PREDICTION

FOR DRUG DISCOVERY WITH SMILES

Introduction

Why AI-powered toxicity prediction is required?

Drug discovery is a field that requires significant amount of wet-lab experiment procedures. Starting wet lab experiments without any base information could be high-cost and considerably risky choice for labs who are searching for a new small molecule as drugs. There, AI-driven drug discovery has been developed through last two decades. One of the key step of AI-driven drug discovery pipeline is toxicity prediction, which can reduce cost in advance of trial stages.

Dataset Selection

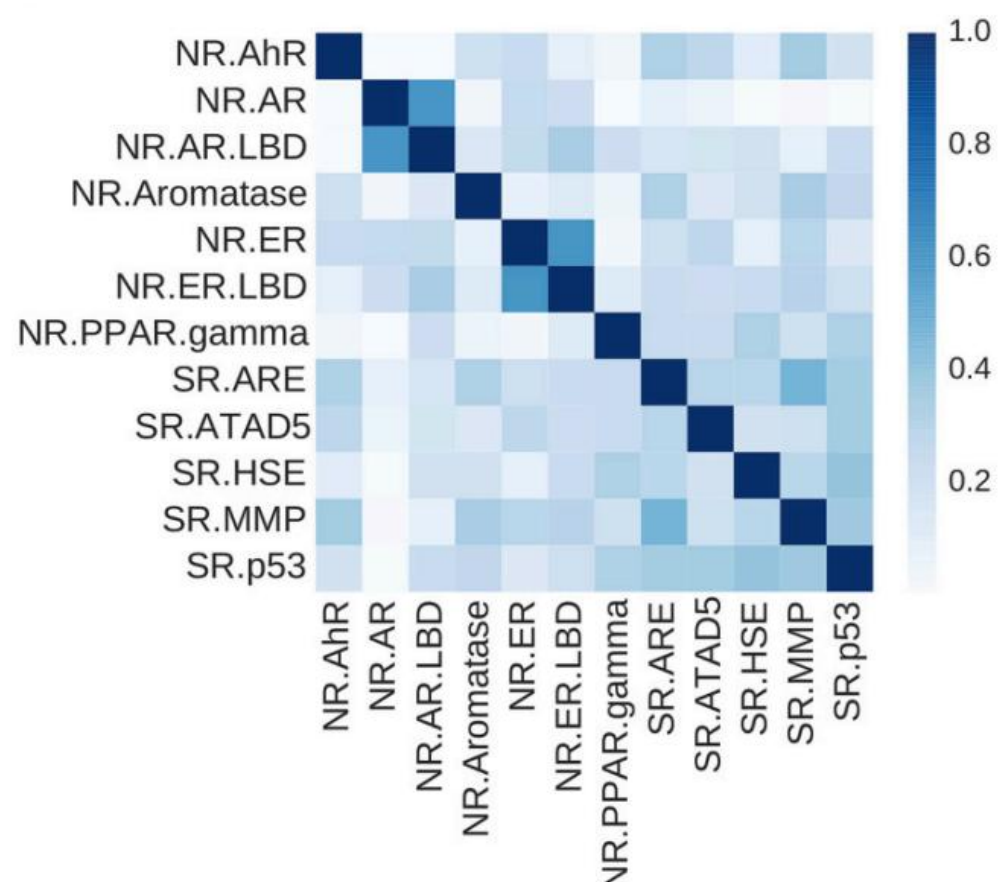
Clintox dataset: labeled with FDA approval

Chemical compound expressed as their SMILES representations. Labeled according to the FDA approval result from the very final stage of the clinical trial. Total 1480 datapoints in raw dataset.

	smiles	FDA_APPROVED	CT_TOX
0	<chem>*C(=O)[C@H](CCCCNC(=O)OCCOC)NC(=O)OCCOC</chem>	1	0
1	<chem>[C@@H]1([C@@H]([C@@H]([C@H]([C@@H]([C@@H]1Cl)C...</chem>	1	0
2	<chem>[C@H]([C@@H]([C@@H](C(=O)[O-])O)O)([C@H](C(=O)...</chem>	1	0
3	<chem>[H]/[NH+]=C(/C1=CC(=O)/C(=C\C=c2ccc(=C([NH3+])...</chem>	1	0
4	<chem>[H]/[NH+]=C(\N)/c1ccc(cc1)OCCCCCOc2ccc(cc2)/C(...</chem>	1	0
5	<chem>[N+](=O)([O-])[O-]</chem>	1	0

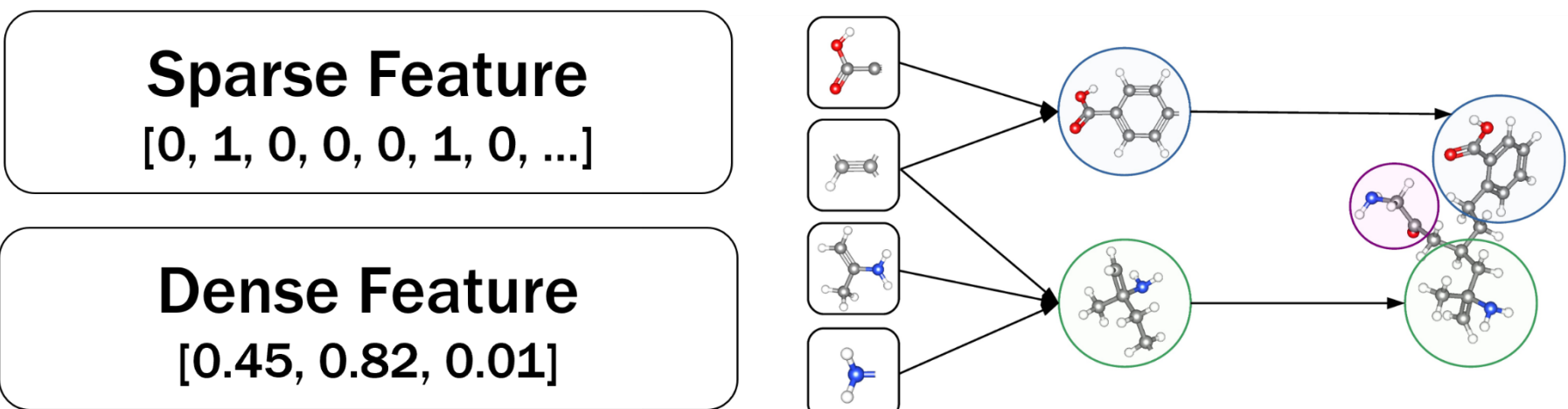
Tox21 dataset: aimed find a new toxicological assessment method

Chemical compounds evaluated across 12 (NR/SR) toxicological assays to develop new methods for toxicity testing. 12 assays are weakly correlated each other. Total 1907 datapoints in raw dataset.



DeepTox project: successful benchmark using Tox21

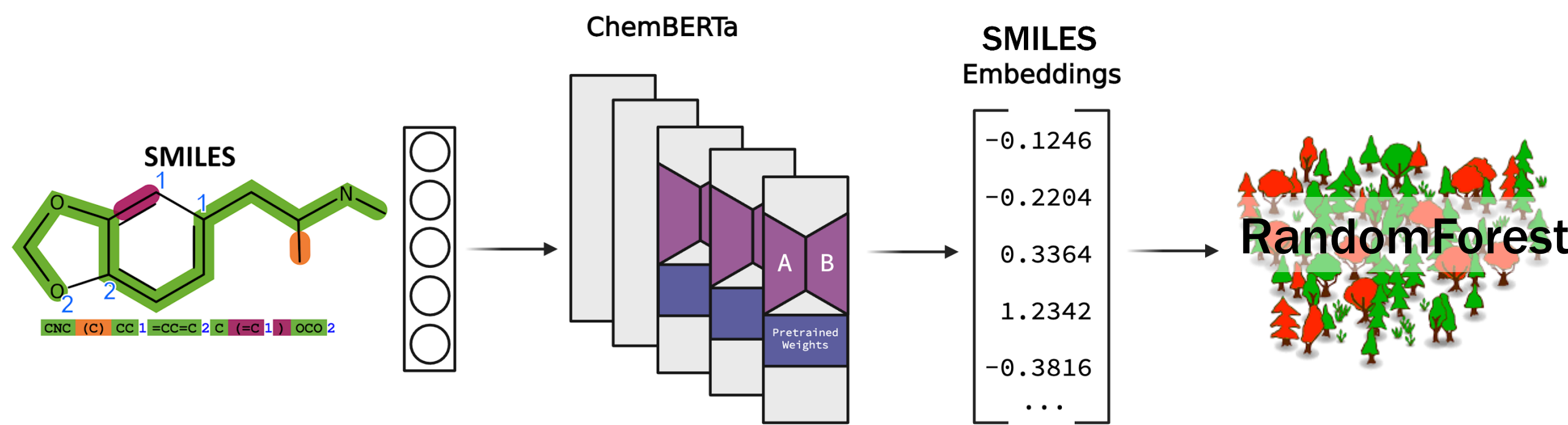
With dense features and sparse features extracted from the Tox21 dataset, DeepTox project successfully trained multi-task toxicity prediction model. Its baseline is Deep Random Forest model which combines deep neural network (DNN) and random forest model.



Method

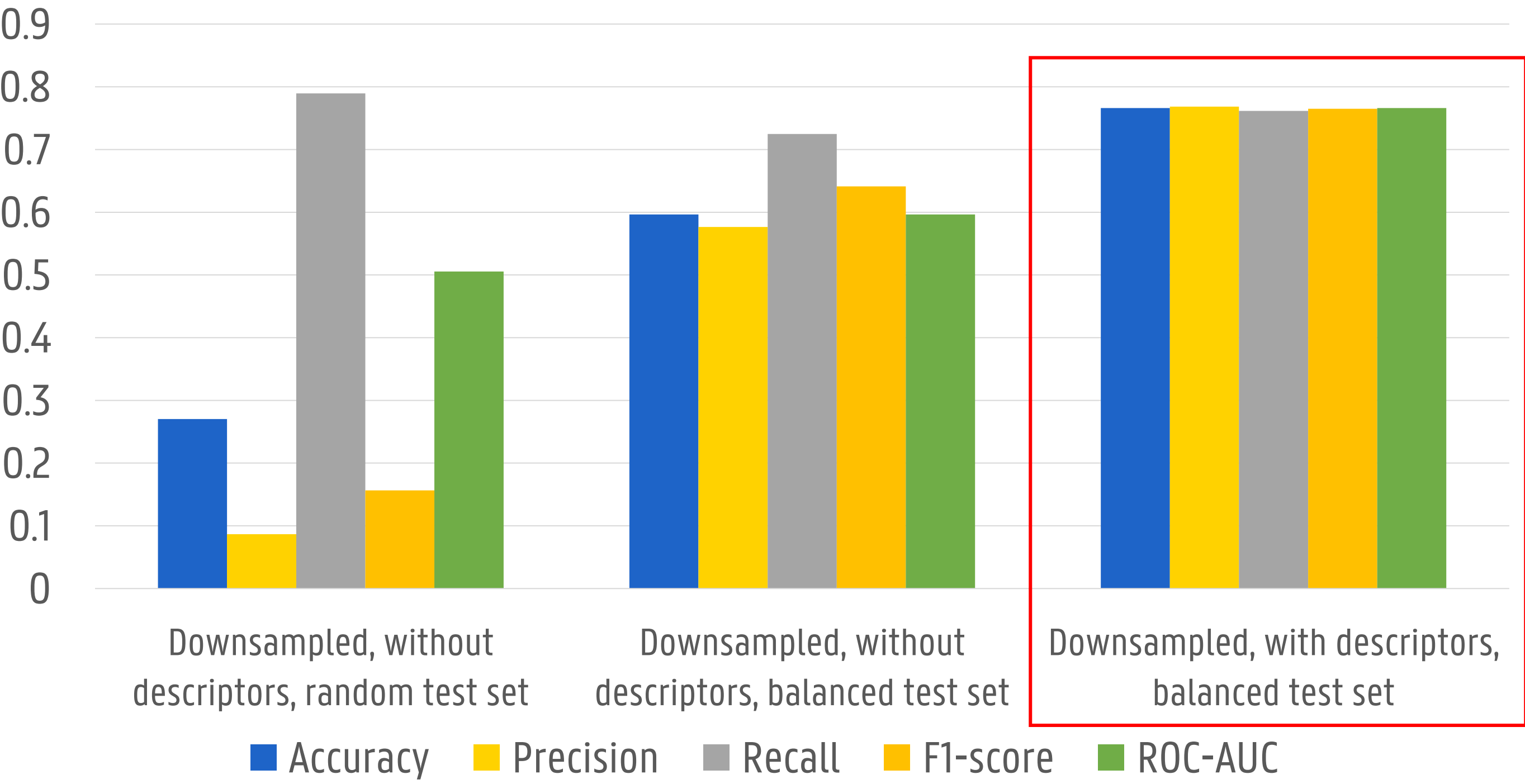
Model Baseline

ChemBERTa, a pretrained large language model (LLM) model with chemical molecules (SMILES) tokenize SMILES to make embedding features to put as an input of RandomForest (RF) model.

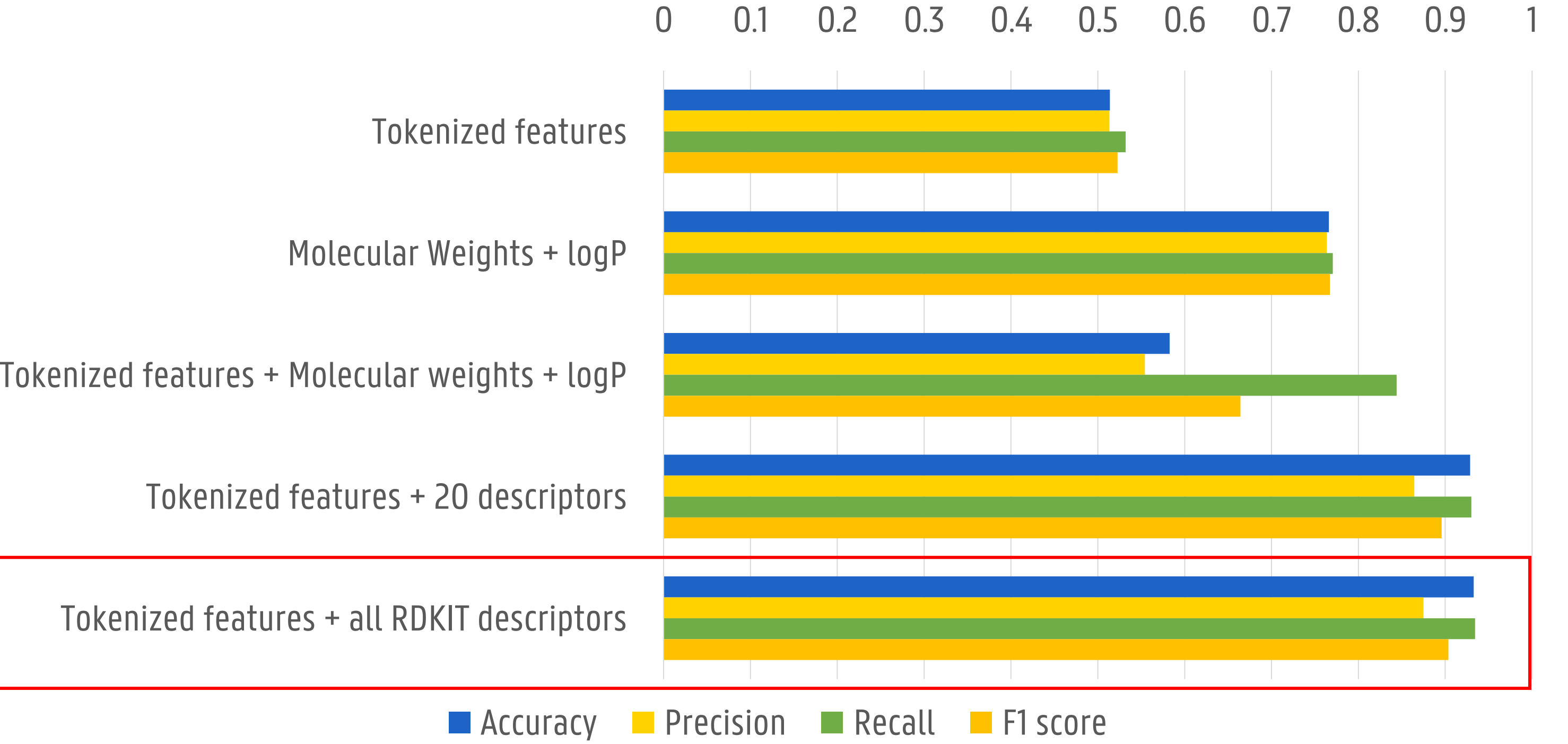


Result

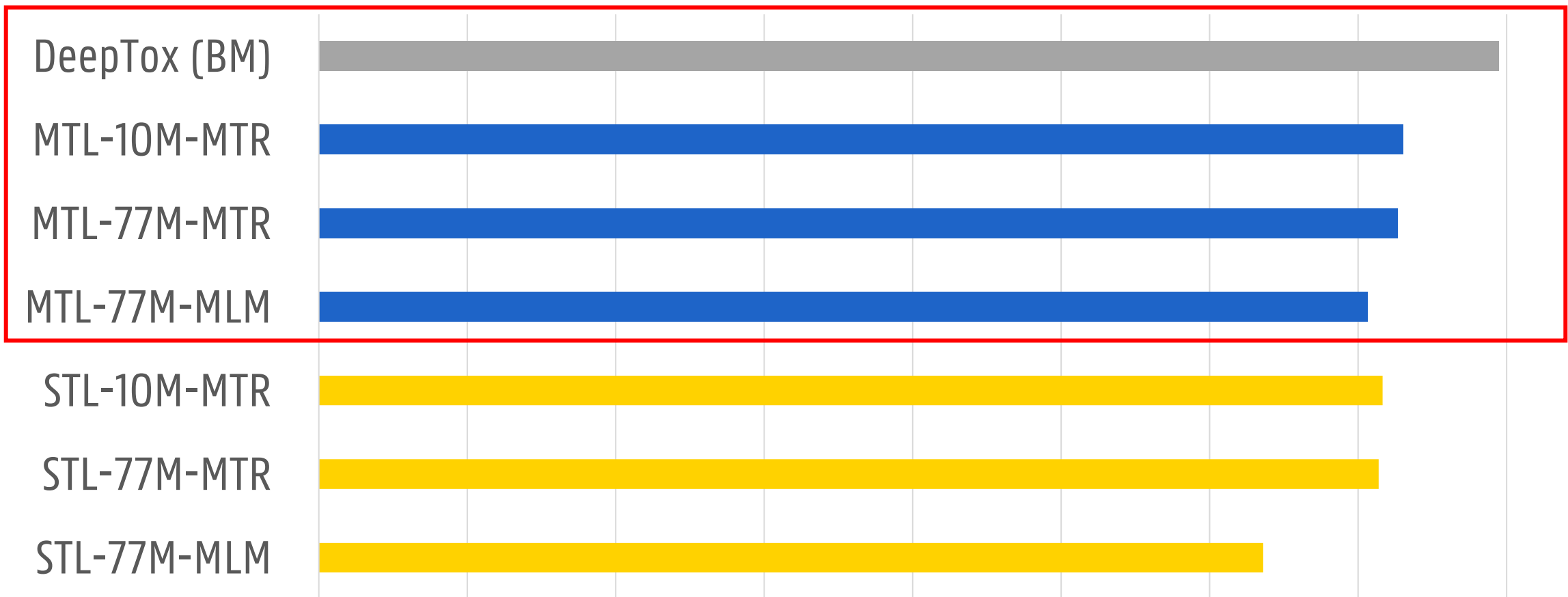
Clintox: Model Performances according to the training settings



Clintox: importance of RDKit descriptor features



Tox21: Multi-task learning (MTL) vs Single-task learning (STL), MTR vs MLM



Besides the benchmark model (DeepTox-MTL), according to their ROC-AUC score, MTL-10M-MTR outperformed on NR-AhR, NR-ER, SR-HSE, SR-MMP. STL-77M-MTR outperformed on NR-AR, NR-AR-LBD, even better performance than the benchmark model. Larger pretraining dataset did not assure the performance since domain specific fine tuning on toxicity prediction task was not done before training with the new dataset. STL worked better than MTL in those tasks that have negligible correlation with other tasks. In those cases, other features from different tasks interrupted the model to give correct and focused prediction in each specific task that has low correlation with others.

In most cases, MTR worked better than MLM, and it indicates that pretraining with regression is more efficient in toxicity prediction. This suggests to new studies; domain specific finetuning and MTL optimization on toxicological assessment field.

Discussion

Clintox

BalanJ descriptor has the highest impact on result because it measures the complexity of the chemical molecule, and more complex more unexpected in human body. Downsampling works better than other settings to resolve the data class unbalance problem.

Tox21

MTL works better than STL since TOX21 tasks are correlated so MTL helped mitigate class imbalance problem by sharing useful information across tasks. For pretraining method, MTR gives better performance than MLM in STL, and no big difference in MTL.

Future Study Plans

- Undersampling vs Oversampling vs Data augmentation
- Best dataset combination for toxicity prediction (Clintox/Tox21/dense/sparse ...)
- LLM with GNN: graph-based toxicity prediction (classification) using tokenized SMILES
- Transfer Learning with Domain-Specific Models
- Multi-modal Toxicity Prediction Models
- Explainable Toxicity Prediction Models

References

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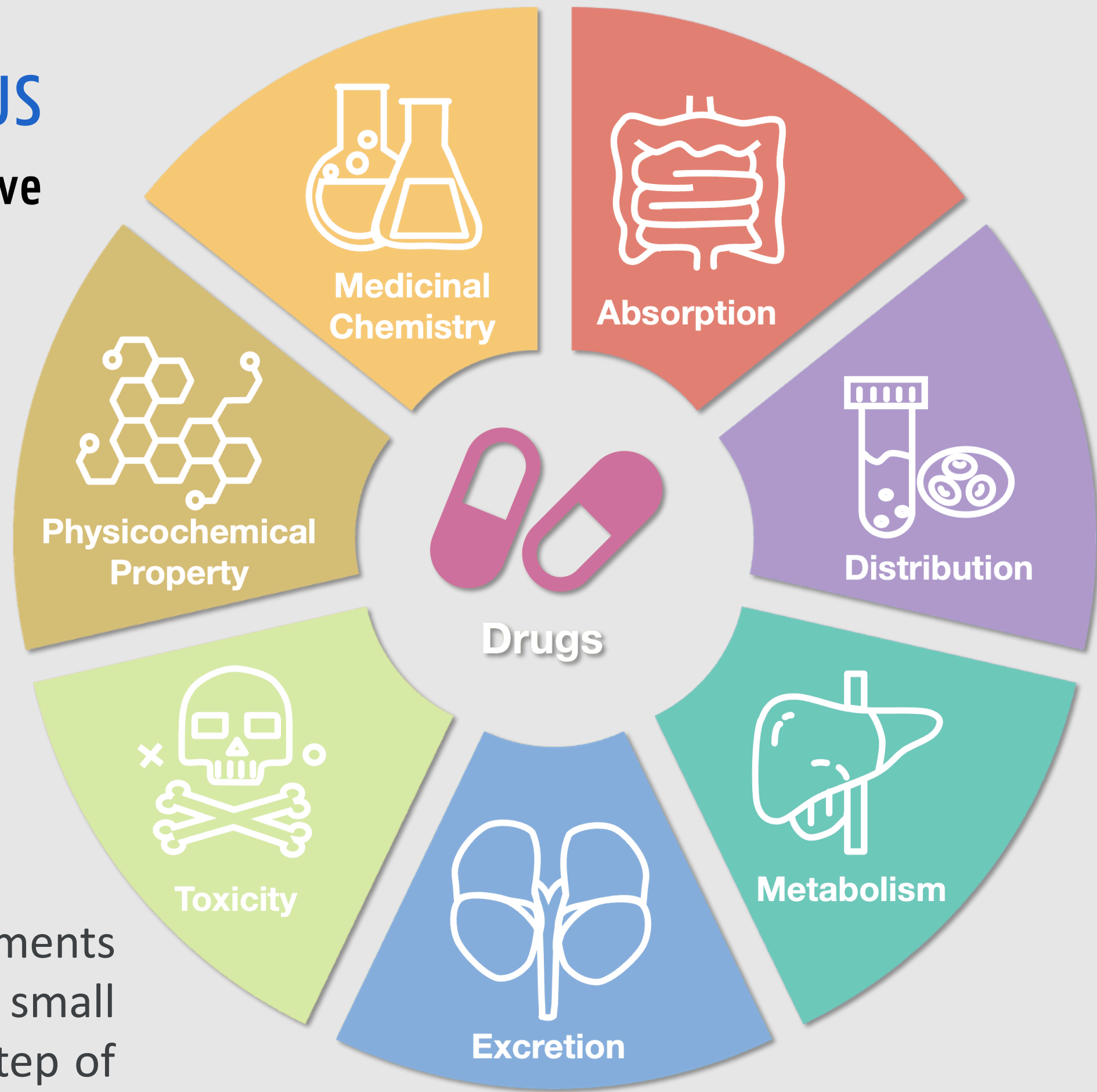
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PPT slides

Kaggle Notebook